



Semester 1 Examinations 2012 / 2013

Exam Code(s) 1SD1/1SD3/1MF1/1IT1
Exam(s) Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development, Master of Information Technology

Module Code(s) CT874/CT536
Module(s) Programming I/Object Oriented Programming I

Paper No. I

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Instructions: Candidates are required to answer:

Section A (20 Marks) Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted. **Answers are to be written on the MCQ sheet provided, NOT on the Examination paper.**

Section B (80 Marks) Answer Any **Two** Questions.

Duration 2 hours
No. of Pages 6
Department(s) Information Technology

Requirements MCQ

Section A (20 Marks)
Attempt all 10 questions
Mark your answers on the MCQ sheet provided

- 1) Polymorphism allows us to _____.
 - a) have the same message execute a different method depending on the receiving object
 - b) hide the internal workings of a Java program from the client programmer
 - c) overload methods and allow the same method name to be used multiple times
 - d) extend and modify code
- 2) How would the three classes, Undergraduate, Graduate, and Student interrelate if we are determined to use inheritance in the class definitions?
 - a) Student would be a subclass of Undergraduate, which would be a subclass of Graduate.
 - b) Graduate would be a subclass of Undergraduate, which would be a subclass of Student.
 - c) Graduate would be a subclass of Student, and Undergraduate would be a subclass of Graduate.
 - d) Undergraduate and Graduate would both be subclasses of Student.
- 3) When an array is passed to a method _____.
 - a) only the reference is passed, and a copy of the array is not created in the method.
 - b) a copy of the array is created within the method.
 - c) the value of the array is passed into the method.
 - d) None of the above.
- 4) The practice of writing multiple constructors to handle different sets of inputs is known as _____.
 - a) inheritance
 - b) polyinputting
 - c) overloading
 - d) multiplicity
- 5) What does the call, "super()" do?
 - a) It will do nothing.
 - b) Call the subclass's constructor
 - c) Call the superclass's constructor.
 - d) This is an invalid call.

- 6) Why would a user want to create multiple constructors?
- a) Multiple constructors are easier to write than a single constructor.
 - b) The Java compiler expects multiple constructors.
 - c) Multiple constructors give the client more flexibility in creating instances.
 - d) Programs with multiple constructors are easier to read.
- 7) When an exception occurs, the normal sequence of flow is terminated and the exception is _____.
- a) accessed
 - b) thrown
 - c) terminated
 - d) initialized
- 8) What is meant by “returning an object” from a method?
- a) It means you are returning a null value to the caller.
 - b) It means returning a primitive type to the caller.
 - c) It means returning the object’s address to the caller.
 - d) It means returning the object’s value to the caller.
- 9) What is the difference between a sentinel-controlled loop and a count-controlled loop?
- a) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed infinitely.
 - b) A count-controlled loop uses the numerical increment operator ++, while the sentinel-controlled loop uses Boolean expressions.
 - c) A count-controlled loop is executed for a fixed number of times while a sentinel-controlled loop is executed repeatedly until a certain value is encountered.
 - d) A count-controlled loop does not take user input in its body, while a sentinel-controlled loop does.
- 10) Which of the following is not a Boolean operator?
- a) &&
 - b) !
 - c) ||
 - d) ++

Section B (80 Marks)
Answer Any TWO Questions

Q 2.

- a) In relation to object-oriented programming with Java, detail your understanding, using examples/analogies where appropriate, of each of the following:

- i. Encapsulation,
- ii. Overloaded Constructors
- iii. Inheritance
- iv. Pass by reference and pass by value.

[15]

- b) Hill's Toyshop require you to write a Java application which allows the user to input 7 daily sales figures, one for each day of the week. The application should identify the highest sales figure for the week and identify which day of the week it occurred. The application should also show how much each day's sales figure differs from the highest sales figure.

[25]

Q 3.

- a) In relation to the visibility of data members, discuss the effects of the modifiers *protected*, *public* and *private*, using examples where appropriate.

[10]

- b) Write a program that replies either *Leap Year* or *Not a Leap Year* when given a year. It is generally a leap year if the year is divisible by 4 but not by 100 (i.e. 1796 is a leap year because it is divisible by 4 but not by 100). However, a year that is divisible by both 4 and 100 is a leap year if it is also divisible by 400 (i.e. 2000 is a leap year but 1800 is not).

You answer should consists of **two** files, a programmer defined class called **LeapYear**, which can determine whether or not a given year is a leap year and an application class called **IsLeapYear**, which uses the reusable class LeapYear to determine whether a given year is a leap year or not.

[30]

Q 4.

ACME University, have requested the development of an application to collect exam result information for each student sitting exams. The application should allow the user to enter the name, ID number and exam results (which lie in the range of 0-100) for each student.

The student population is made up of both undergraduate and postgraduate students and students are each required to sit five exams. The calculation of a pass mark differs for both undergraduate and postgraduate students. Undergraduate Students pass if the average mark of their 5 exam results is greater than or equal to 50. Postgraduate students pass if the average mark over the 5 exams is 70 or above.

The university wishes to maintain all of this information in **one** collection.

You are required to write an application to display the name and ID number of each student and to calculate whether they have passed or failed. The application should contain the following:

- A **Student** class with appropriate overloaded constructors. The class should contain private fields with public methods to change and retrieve the values of private instance variables. You should also implement a CalculateResult method, which can be overwritten.

[10]

- An **Undergraduate** class with a CalculateResult method and appropriate overloaded constructors to assist with input.

[5]

- A **Postgraduate** Class with a CalculateResult method and appropriate overloaded constructors to assist with input.

[5]

- A **StudentTest** application class which implements a collection of five students containing a mix of both postgraduate and undergraduate students and display their names, ID numbers and whether they have passed or failed.

[20]

Q 5.

- a) What is Polymorphic behavior? Discuss the advantages of polymorphism as a programming strategy, using examples where appropriate.

[10]

- b) Write a Java application that allows the user to input:

1. text
2. a character to search for

The application should display the number of times the search character appears in the text.

[10]

- c) Write a Java application that reads in a number of tokens from the user and displays the tokens in reverse order.

[10]

- d) Write a method *findMinimum* that returns the smallest of three floating-point numbers. Use the `Math.min` method to implement *findMinimum*. Incorporate the method into an application that reads in three numbers from the user, determines the smallest value and displays the result.

[10]